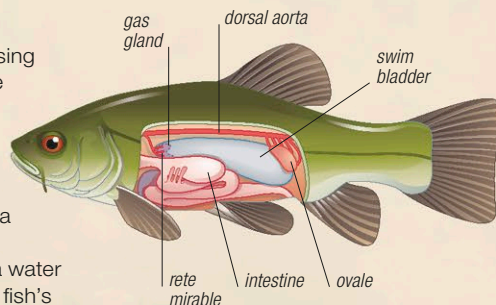


Buoyancy

Most bony fishes control their buoyancy using an outgrowth of the intestine known as the swim bladder. The volume of gas in the swim bladder is affected by water pressure (depth), but the fish can maintain correct buoyancy by adding and subtracting gas. The swim bladder of a freshwater fish has a greater capacity than that of a marine fish, because freshwater is less dense than sea water and does not give as much support to the fish's body. Having a swim bladder means that bony fishes (unlike cartilaginous fishes) do not need to use their fins for buoyancy control and, so have been free to modify them for many other uses.



SWIM BLADDER

Apart from some primitive species, most bony fishes adjust their buoyancy by moving gas between the bloodstream and the swim bladder.

Gas enters the bladder through the gas gland, which is supplied with blood by a network of capillaries called the rete mirabile.

Using another structure, known as the ovale, gas is removed from the bladder and reabsorbed into the blood.

Reproduction

In most bony fishes, fertilization occurs outside the body but in a few species it is internal. Marine fishes that live in open water usually produce eggs that float as part of the plankton, often in large numbers because only a small number will survive. Coastal marine and freshwater fishes tend to produce fewer eggs, which are laid in the sediment, in nests, or attached to plants. A few species care for their eggs or even their young. This usually takes the form of protecting them from predators. Bony fishes include some hermaphroditic species, which have both male and female reproductive organs. Others change sex during the course of their adult life, either from male to female or vice versa. These alternative strategies are all designed to maximize

reproductive output using minimum energy.



PARENTAL CARE

The male yellowhead jawfish incubates its eggs by storing them inside its mouth.

Some other bony fishes (including some cichlids) use their mouth to protect their young from predators.



LARVAE

Most bony fishes hatch from eggs and begin life as larvae. These larvae of the European eel are in the process of metamorphosis, during which they are referred to as glass eels.

Schooling

Many bony fishes swim in large groups that often contain tens of thousands of individuals. When the fishes are grouped together but behave independently, the group is referred to as an aggregate. In a school, however, the movement of the fishes is much more harmonized, to the extent that the school resembles a single organism. Only bony fishes can swim in this coordinated way. Each fish usually follows a course parallel to its neighbors, maintaining its position using its vision, hearing, and lateral-line system. Fishes in a school are safer than solitary ones, because there are more individuals to look out for danger and it is difficult for a predator to pick out a single fish from a large mass (see left). It is also easier for fishes in a group to find mates and locate food.



STAYING TOGETHER

Schooling fishes pack tightly together, each one typically keeping a distance to its nearest neighbor of no more than one body length. A silvery coloration (seen in these sweepers) helps the fishes see each other's movements because a small change in direction produces a large change in the amount of light reflected.